BESS — 40ft 3.5 MWh containerised

Revision Rev A · Fractional Forge · In build

Forge project pack

Project name	BESS — 40ft 3.5 MWh containerised
Revision	Rev Rev A
Project created	25 Apr 2026, 05:42
Brief locked	25 Apr 2026, 18:30
Shipped	Not shipped yet
Document generated	4 May 2026, 12:43

RED **Feasibility — blockers found**

- Design does not fit the declared envelope.
- Estimated unit cost is 2.8× the brief ceiling.

Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

BRIEF INFEASIBLE — DO NOT PROCEED TO PROCUREMENT

The design as currently briefed cannot be built within the declared envelope, mass, or transport constraints. Modules, bill of materials, cost waterfall, reconciliation, risks register and supplier shortlist are intentionally omitted from this document. Resolve the blockers listed on the Feasibility Exception page before any module decomposition or supplier engagement.

TOTALS AT A GLANCE

MODULES	KEY PARTS	BOM ROWS	SUPPLIERS
9	**72**	**80**	**26**

FAILURE MODES	OPEN QUESTIONS	STANDARDS	REVIEWS
37	**27**	**11**	**3**

UNIT COST	CEILING	HEADROOM
— Not computed — design infeasible	**—**	**—**

Contents

Feasibility

Before this report was assembled, the design was checked against the brief constraints. The verdict below is computed deterministically from the sizing solver, bill of materials, and cost waterfall — not from language-model opinion.

RED **Feasibility — blockers found**

- Design does not fit the declared envelope.
- Estimated unit cost is 2.8× the brief ceiling.

1. Brief

'LLM | GEMINI-3.1-PRO

RED **Feasibility — blockers found**

- Design does not fit the declared envelope.
- Estimated unit cost is 2.8× the brief ceiling.

WHAT WE ARE BUILDING

A 40-foot containerised battery energy storage system delivering 3.5 MWh and 1.5 MW for behind-the-meter commercial and industrial sites in the United Kingdom. Lithium iron phosphate cells, integrated battery management, bi-directional power conversion with grid-forming capability, liquid-cooled thermal management, and certified fire detection and suppression. Target installed capital cost under 180,000 pounds. Must ship complete on a flatbed lorry, install on a prepared concrete pad, and commission inside two days. Compliant with United Kingdom G99 grid code and the latest National Fire Chiefs Council guidance for lithium battery storage.

MISSION

Deliver a factory-integrated, containerised lithium iron phosphate battery energy storage system that enables UK commercial and industrial sites to optimise energy costs and support grid stability behind the meter, without specialist on-site build work.

USE CASE

A prepared-pad, two-day-commission containerised BESS that ships complete on a flatbed lorry and connects behind the meter to reduce peak demand charges and provide grid services for UK C&I sites.

TARGET CUSTOMERS

UK commercial and industrial site operators, energy-intensive manufacturers, logistics and distribution centres, property developers with large commercial tenants

WHY NOW

UK grid code G99 now mandates grid-forming capability for new behind-the-meter storage connections; NFCC guidance on lithium battery storage has matured into a de-facto planning requirement; LFP cell costs have fallen sufficiently to make containerised C&I BESS commercially viable at scale; DNOs are actively incentivising demand-side flexibility.

Constraints declared

Unit cost ceiling	—
Max mass	40,000.00 kg

SECTION JUDGEMENT — COMPOSITE SCORE: 7.0/10

Factual Integrity	**8/10**	Constraint data aligns well with realistic physical limits for a 40ft container.
Visual Clarity	**5/10**	Heavy text blocks make scanning constraints difficult without visual separation.
Linguistic Precision	**7/10**	Terminology is industry-standard, though occasionally verbose.
Logical Coherence	**8/10**	Mission, use case, and rationale follow a solid deductive flow.
Actionability	**7/10**	Provides clear targets but lacks explicit constraint checklists.

What is good: Narrative strength is excellent. The constraints provide solid boundary definitions.

What is bad: Walls of text create cognitive friction. The layout does not visually separate the 'Mission' from technical 'Constraints'.

Next-round coding improvements by Gemini:

I, Gemini, will implement a structured Key-Value pair format. Constraints should be a bulleted checklist rather than paragraphs.

2. Regulatory posture

ÁDB

Per-standard compliance matrix — applicability, design impact, evidence required, status, owner, and the next concrete gap action.

5 of 5 regulatory entries are unverified extractions.

These entries were populated by an automated extraction pass and should be cross-checked against the original standard text before any procurement, certification, or design-freeze decision.

ENA G99 NOT STARTED Electrical lead Unverified

Requirements for the Connection of Generation Equipment in Parallel with Public Distribution Networks

Mandatory UK grid code for behind-the-meter storage connecting in parallel with the distribution network; grid-forming capability required.

Applicability: Applicable to the 1.5MW grid-scale inverter.

Design impact: Requires SiC or advanced IGBT bi-directional inverter.

Evidence required: Type test certificate or individual commissioning test.

Next action: Verify SMA inverter compliance with latest G99 amendment.

BS EN IEC 62619 NOT STARTED Battery safety lead Unverified

Secondary Lithium Cells and Batteries — Safety Requirements for Use in Industrial Applications

Cell and module-level safety standard required by UL 9540 for LFP battery components

Applicability: UK statutory + EU — every cell + module supplied into the BESS must meet BS EN IEC 62619 testing for industrial deployment.

Design impact: Cell vendor selection limited to those with current 62619 certificates (CATL/EVE/Sunwoda); module-level abuse testing at supplier.

Evidence required: BS EN IEC 62619 test report from accredited lab (TÜV / DEKRA / Intertek) per cell SKU.

Next action: Add 62619 cert demand to the cell-supplier RFQ.

BS EN IEC 63056 NOT STARTED Battery safety lead Unverified

Safety requirements for secondary lithium cells and batteries for use in electrical energy storage systems

System-level safety certification required for the complete BESS installation.

Applicability: UK statutory + EU — system-level safety standard for stationary BESS.

Design impact: BMS architecture, contactor + fuse coordination, thermal runaway propagation barriers, enclosure venting design.

Evidence required: BS EN IEC 63056 test report from accredited lab; system-level Declaration of Conformity.

Next action: Book accredited-lab campaign 6+ months before commissioning.

UL 9540A NOT STARTED Battery safety lead Unverified

Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems

Thermal-runaway propagation test recommended to demonstrate fire safety in support of NFCC guidance compliance.

Applicability: International (US-origin) — NFCC + UK fire authorities increasingly demand UL 9540A test data.

Design impact: Drives module-level barrier specification (mica/aerogel placement); enclosure vent area sizing; deflagration vent panel design.

Evidence required: UL 9540A test report at cell, module, unit AND installation levels.

Next action: Source cells whose vendor already has unit-level UL 9540A.

NFPA 855 NOT STARTED Product safety manager **Unverified**

Standard for the Installation of Stationary Energy Storage Systems

Governs minimum fire separation distances, deflagration venting, and suppression requirements.

Applicability: International (US-origin) but adopted as a de-facto reference by UK NFCC.

Design impact: Fire separation: 3 m to property lines; deflagration vent area calc per NFPA 68; gas-suppression agent (Novec 1230) sizing.

Evidence required: Fire safety strategy referencing NFPA 855 clauses; vent-area calc.

Next action: Engage a fire-safety consultant 3 months before planning submission.

SECTION JUDGEMENT — COMPOSITE SCORE: 8.0/10

Factual Integrity	**9/10**	Citations of G99, NFPA 855, and UL 9540A are perfectly aligned with UK battery storage standards.
Visual Clarity	**6/10**	Tables are prone to breaking across pages, cutting vital compliance information in half.
Linguistic Precision	**8/10**	Action items ('Verify SMA inverter compliance') are clearly articulated.
Logical Coherence	**9/10**	Categorisation by code, summary, impact, and action logically flows from risk to resolution.
Actionability	**8/10**	Gap actions provide immediately usable tasks for engineering leads.

What is good: The compliance matrix structure is solid, and the unverified extraction warnings add critical trust.

What is bad: React-PDF struggles with large tables across page breaks. Rows can clip in half if not managed.

Next-round coding improvements by Gemini:

I, Gemini, will ensure strict wrap={false} on table row <View> components so they break cleanly to the next page.

3.1 · Container Enclosure & Structure

ÁDB

5200.0 kg / budget 5000.0 kg · 10 wk lead · Specialist judgement

Container Enclosure & Structure

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Modified 40-ft ISO high-cube steel container providing the weatherproof, fire-rated, structural housing for all subsystems with service aisle, doors, roof penetrations, and ISO corner castings.

WHY IT MATTERS

The container is the primary structural, environmental, and fire-safety boundary of the entire product — it carries every subsystem load, defines IP/weather rating, provides the NFPA 855 fire compartment, and is the single earth reference for the whole system.

DESCRIPTION

The enclosure is a new-build (rather than second-hand refurbished) 40-ft ISO 1AAA high-cube steel container, modified at the fabricator to suit BESS duty: reinforced undercarriage for ~3,000 kg/m^2 floor loading, pre-cut and framed roof penetrations for the ceiling HVAC unit and deflagration vents, two opposing personnel doors with panic hardware, and an internal fire-rated rockwool/steel sandwich lining giving a 1-hour rating per NFPA 855 Chapter 9.

KEY PARTS (7)

- Modified 40-ft ISO high-cube container shell, external 12,192 × 2,438 × 2,896 mm
- Reinforced steel deck floor replacing standard 28 mm plywood
- Internal thermal/fire-rated lining: 50–75 mm rockwool with 1-hour fire-rated inner steel sheet
- Personnel service door 900 × 2,100 mm single-leaf outward-opening with crash/panic bar
- Roof penetration frames for HVAC unit (~2,000 × 1,000 mm cut-out)
- Single-point earth bonding bar (copper, 40 × 5 mm)
- Fire-rated sealed cable transit penetrations (Roxtec-type, 2-hour rated)

FAILURE MODES (4)

- Floor deflection or weld cracking under concentrated battery rack loads
- Corrosion perforation of Corten walls or roof around HVAC and cable penetrations
- Fire-rated lining degradation or seam failure compromising the 1-hour compartment rating
- Loss of earth bonding continuity at rack/PCS interfaces causing touch-voltage hazard

OPEN QUESTIONS (3)

- Whether deflagration venting via roof panels or forced mechanical ventilation is preferred
- Final floor-loading spec: is 3,000 kg/m^2 sufficient
- Whether the step-up transformer is internal or external

Estimated cost	**£19,050**

ENGINEERING REVIEW (1)

Manufacturing review · WARNING

The structural envelope is fundamentally sound - ISO container construction is a mature, globally-manufactured form. However, specific issues require resolution before this module is production-ready. Two of them - the coating dry film thickness and the internal clear-width interface - will cause quality escapes at first article if not addressed now.

[WARNING] Manufacturing: Coating specification inadequate.

Revise coating specification to 280 µm target / 240 µm minimum dry film thickness.

[CRITICAL] Assembly: Internal clear-width interface unconfirmed.

Lock battery rack maximum width to d 2,320 mm and document as a hard interface constraint.

Recommendations

- Place a hard milestone for UL 9540A data four weeks ahead of container fabrication start.
- Fix wall panel gauge at 2.0 mm across all production units.
- Add corner casting weld qualification requirements to the specification.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.2 · Port-side LFP Battery Rack Array

ÁDB

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement

Port-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses half of the LFP battery modules along one long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

This module represents half of the product's stored energy and therefore half of the project's bill-of-materials cost; its cell chemistry, module format, and rack architecture drive the system's round-trip efficiency, cycle life, warranty exposure, and UL 9540A fire-propagation behaviour.

DESCRIPTION

The port-side battery rack array comprises two free-standing steel-framed racks arranged along the port long sidewall of the container, each housing 10-12 LFP prismatic-cell modules stacked vertically and interconnected by copper busbars to form a ~500 kWh / ~1500 V DC nominal string. Each rack integrates a rack-level BMU monitoring per-cell voltage and per-module temperature, a main DC contactor with pre-charge resistor and NH-style fuse for string isolation, and an insulation-monitoring tap.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost	**£76,000**

ENGINEERING REVIEW (1)

Manufacturing review · WARNING

The rack architecture is sound and buildable with standard sheet-metal fabrication, but the Specify page is missing the fields that drive tooling and supplier selection.

> **[WARNING] Manufacturing: No target material/process/tolerance specified on the module card.**
> Fill the module card — target material (CR steel / S275-class), process (laser+brake+MIG+powder coat), tolerance (ISO 2768-mK).

> **[CRITICAL] Assembly: Seismic load path through the container corrugation is under-engineered.**
> Redesign the seismic load path — welded C-channel stringer off the container's structural top/bottom rails.

Recommendations

- Belleville washers + torque-stripe witness marks on every busbar joint.
- Clearance/creepage diagram for 1,500 V DC class-II
- Standardise fasteners to M12×40 8.8 zinc

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	**9/10**	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	**8/10**	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	**9/10**	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	**9/10**	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	**10/10**	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.3 · Starboard-side LFP Battery Rack Array

ÁDB

2400.0 kg / budget 2500.0 kg · 16 wk lead · Specialist judgement · mirrors Port-side LFP Battery Rack Array

Starboard-side LFP Battery Rack Array

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Houses the mirrored half of the LFP battery modules along the opposite long sidewall, providing DC energy storage with integrated rack-level BMS.

WHY IT MATTERS

Together with the port-side array, this module stores 100% of the system's usable energy; its reliability, thermal uniformity, and safe failure behaviour directly determine round-trip efficiency, warranty cycle life, and most critically the probability of a thermal runaway event.

DESCRIPTION

The starboard-side battery rack array is the mirrored counterpart to the port-side array, comprising 2 floor-standing LFP battery racks arranged along the starboard long sidewall of the container, separated from the central service aisle by ~1,000 mm clear working space.

KEY PARTS (7)

- 2x LFP battery racks, 600 x 1200 x 2200 mm
- 20-24x LFP prismatic-cell modules (10-12 per rack)
- Rack-level BMU (Battery Management Unit) per rack
- Rack main DC contactor and pre-charge circuit, 1500 V DC / 400 A rated
- Intra-rack DC busbars, tin-plated copper, 30 x 10 mm cross-section
- Rack top air plenum with filtered supply-air inlet
- Seismic bracing brackets, 5 mm steel

FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Estimated cost **£76,100**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	**9/10**	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	**8/10**	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	**9/10**	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	**9/10**	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	**10/10**	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.4 · DC Combiner & Distribution Panel

ÁDB

180.0 kg / budget 200.0 kg · 10 wk lead · Specialist judgement

DC Combiner & Distribution Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Aggregates, fuses, and protects the DC outputs of all battery racks and routes combined DC to the PCS.

WHY IT MATTERS

The DC combiner is the single aggregation point where all battery string energy converges; it is the last line of defence before a fault propagates to the PCS or back into healthy battery racks.

DESCRIPTION

The DC Combiner & Distribution Panel is a floor-standing, IP54-rated steel cabinet (approx. 600 × 400 × 1800 mm) located on the short-end wall of the container adjacent to the PCS inverter. It aggregates the DC outputs of the 4–5 LFP battery racks onto a common tinned-copper busbar, then routes the combined DC bus via a single high-current cable pair to the PCS.

KEY PARTS (9)

- IP54 floor-standing steel enclosure, 600 × 400 × 1800 mm
- DC fuse holders with NH2 gPV fuses, 1500 V DC rated
- Main DC contactor, 1500 V DC / 1000 A
- DC arc-fault circuit interrupter (AFCI) module
- Copper busbar, tinned, 80 × 10 mm cross-section
- Hall-effect DC current transducer (LEM DHAB)
- Surge protection device (SPD) Type II
- Insulation monitoring device (IMD)
- Auxiliary 24 V DC control terminal block

FAILURE MODES (4)

- Fuse nuisance trip or coordination mis-match causing loss of a battery string
- DC contactor weld-closed failure preventing emergency isolation
- Insulation resistance degradation from moisture ingress or cable chafing
- Busbar hot-spot due to loose torque on bolted joints

OPEN QUESTIONS (3)

- Final number of rack feeders (4 vs 5) pending battery module/rack vendor selection
- Whether the PCS vendor requires integrated pre-charge circuitry inside the combiner
- DNO/insurer preference on grounded vs ungrounded (IT) DC topology

Estimated cost **£10,890**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

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Next-round coding improvements by Gemini:

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3.5 · Bidirectional PCS Inverter

ÁDB

1200.0 kg / budget 1300.0 kg · 16 wk lead · Specialist judgement

Bidirectional PCS Inverter

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Converts bidirectionally between DC battery bus and 400 V AC at 500 kW for charge/discharge operation.

WHY IT MATTERS

The PCS is the single point of energy conversion between the battery and the grid and therefore sets the system's round-trip efficiency, grid compliance (G99 / IEEE 1547), response time for peak-shaving dispatch, and fault ride-through behaviour.

DESCRIPTION

The PCS is a floor-standing, cabinet-style bidirectional voltage-source converter rated at 500 kW continuous, converting between the 600–1,500 V DC battery bus and 400 V, 50 Hz, 3-phase AC. It is built around a 3-level IGBT bridge with an LCL output filter, DC pre-charge circuit, AC contactor, and LCL-side EMI filtering, all housed in an IP54 steel enclosure approximately 800 × 800 × 2,000 mm.

KEY PARTS (8)

- 500 kW bidirectional PCS unit
- IGBT power module stack (3-level NPC or T-type topology)
- LCL output filter with copper-wound inductors
- DC-side pre-charge circuit with contactor
- AC output contactor / motorised isolator, 1,000 A
- Digital control card with DSP + FPGA
- Internal forced-air cooling fans
- Integrated auxiliary 24 V DC supply and insulation-monitoring device (IMD)

FAILURE MODES (4)

- IGBT module failure due to thermal cycling or overcurrent
- DC-link capacitor degradation (ESR rise) leading to ripple overheating
- Output LCL filter inductor insulation breakdown
- Cooling fan or air-filter blockage causing thermal derate

OPEN QUESTIONS (3)

- Final DNO-agreed grid code settings (G99 Type A vs Type B)
- Whether a grid-forming (black-start / islanded) capability is required
- Confirmation of DC operating window

Estimated cost	**£113,400**

ENGINEERING REVIEW (1)

Manufacturing review · WARNING

The cabinet itself is a vendor-qualified black box — not our manufacturing problem. But three integration interfaces (air handshake, cable entry, seismic/shipping restraint) will bite us at build rate if we don't specify them now.

[WARNING] Integration: Scope clarity — this is integration, not fabrication.
Split this module into two Specify entries — "PCS Unit (bought-in)" and "PCS Integration Kit (fabricated)".

[CRITICAL] Thermal: Airflow handshake with HVAC return plenum is undefined.
Specify the supply-air boundary condition on the cabinet datasheet.

[WARNING] Assembly: Cable entry and termination consistency.
Pre-fit and pre-terminate all PCS-side cables at the integration facility under torque-witnessed conditions.

Recommendations

- Issue an Interface Control Document covering: bolt pattern, cable gland positions, air inlet/exhaust flange geometry
- Qualify two PCS vendors before committing to the plinth design.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:
I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.6 · AC Switchgear & Step-up Transformer

ÁDB

3200.0 kg / budget 3500.0 kg · 16 wk lead · Specialist judgement

AC Switchgear & Step-up Transformer

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Provides AC protection, metering, G99-compliant grid interface relays, and step-up from LV to MV for the DNO grid connection.

WHY IT MATTERS

This module is the legal and electrical interface between the BESS and the UK distribution network. Without a type-tested G99-compliant protection scheme and correctly-rated transformer, the DNO will not issue connection approval, rendering the entire product unsaleable.

DESCRIPTION

This module comprises the low-voltage AC switchgear cubicle located inside the container adjacent to the PCS, and an externally-skidded step-up distribution transformer sited alongside the container. The LV switchgear is a floor-mounted IP54 steel cubicle (powder-coated RAL 7035) housing the main 800 A MCCB, G99 protection relay, revenue meter, SPDs, auxiliary transformer feeder breaker, and terminal compartments for PCS-side and transformer-side cables.

KEY PARTS (8)

- LV AC switchgear cubicle, 400 V / 3-phase / 50 Hz, 800 × 600 × 2,000 mm, IP54
- G99-compliant protection relay
- Revenue-grade bidirectional energy meter, Class 0.5S
- Step-up distribution transformer, 630 kVA, 400 V Dyn11 ’ 11 kV
- Auxiliary services transformer, 10 kVA, 400 V ’ 230 V
- Surge protection device (SPD), Type 1+2
- LV cabling: 4-core 300 mm^2 XLPE/SWA/LSZH
- Earthing bar, 50 × 10 mm copper

FAILURE MODES (5)

- Protection relay nuisance tripping on grid disturbances
- Transformer winding insulation breakdown from repeated thermal cycling
- MCCB fails to clear a downstream fault within required time
- SPD degradation after lightning/surge events
- Auxiliary transformer overload during simultaneous HVAC + fire-pump operation

OPEN QUESTIONS (3)

- Final DNO connection voltage (11 kV vs 33 kV)
- Whether DNO will accept type-tested relay settings or require site-specific protection study
- Transformer siting: single external skid vs segregated internal compartment

Estimated cost	**£65,330**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.7 · HVAC Thermal Management System

ÁDB

320.0 kg / budget 350.0 kg · 10 wk lead · Specialist judgement

HVAC Thermal Management System

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Ceiling-mounted chiller and ducting that removes heat from battery racks and power electronics to maintain internal temperature envelope.

WHY IT MATTERS

LFP cell life is exponentially sensitive to temperature: sustained operation above 35 degC can halve calendar life and breach the 10-year warranty envelope. The HVAC system is therefore the primary enabler of the warranty, round-trip efficiency, and safe operation.

DESCRIPTION

The HVAC thermal management system is a rooftop-mounted, direct-expansion (DX) packaged chiller arrangement sized to reject the ~25–30 kW continuous heat load generated by the LFP battery racks and PCS at 1C operation, with ~50% thermal margin giving 40–50 kW nominal cooling capacity. Two redundant rooftop units (N+1) mount on reinforced curbs over roof penetrations.

KEY PARTS (9)

- Rooftop packaged DX chiller unit, 45 kW nominal cooling capacity
- Redundant secondary chiller unit, 45 kW (N+1 redundancy)
- Galvanised steel supply air ducting
- Return air plenum with aspirating intake grilles
- Condensate drain system
- Airflow and temperature sensors
- MERV-8 pre-filter and MERV-13 primary filter bank
- HVAC local controller
- Roof penetration curb kits

FAILURE MODES (4)

- Refrigerant leak (microcracks at brazed joints) causing loss of cooling capacity
- Condensate drain blockage or freeze-up leading to water ingress
- Filter clogging increasing fan static pressure
- Compressor failure or contactor welding in a single unit

OPEN QUESTIONS (3)

- Final confirmed heat load from selected cell vendor's UL 9540A data
- Whether UK DNO site ambient design temperature should be 35 degC or 40 degC
- Refrigerant selection lock-in: R454B vs R410A

Estimated cost	**£67,120**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.8 · Fire Detection & Clean-Agent Suppression

ÁDB

320.0 kg / budget 350.0 kg · 10 wk lead · Specialist judgement

Fire Detection & Clean-Agent Suppression

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

NFPA 855 compliant aspirating smoke/off-gas detection, control panel, clean-agent cylinder, and ceiling nozzles for early-warning and suppression of battery fires.

WHY IT MATTERS

Lithium-ion battery fires are the single largest safety, insurance, and permitting risk for containerised BESS. Without NFPA 855 / UL 9540 compliant detection and suppression, the product cannot be insured, cannot obtain UK fire authority and DNO sign-off, and cannot be deployed on commercial sites.

DESCRIPTION

This module provides a two-stage fire safety system for the container: (1) very-early warning via aspirating smoke detection and lithium-ion off-gas sensing (H2, CO, electrolyte VOCs) that triggers alarms and shuts down the PCS/contactors before a thermal runaway propagates, and (2) clean-agent total-flooding suppression using a FK-5-1-12 (Novec 1230) cylinder and ceiling-mounted nozzle network.

KEY PARTS (9)

- Clean-agent cylinder, Novec 1230 (FK-5-1-12)
- Fire suppression control panel
- VESDA-style aspirating smoke detector
- Off-gas / H2 / CO / LEL gas detector array
- Ceiling discharge nozzles
- Schedule 40 seamless steel agent distribution piping
- Manual pull station and abort switch
- Deflagration vent panel on container roof
- Strobe/sonalert alarm devices

FAILURE MODES (4)

- False discharge of clean agent due to single-sensor fault
- Failure to detect early off-gas because sampling tubes become blocked
- Clean-agent flooding fails to prevent re-ignition of a battery in thermal runaway
- Deflagration vent fails to open at design pressure

OPEN QUESTIONS (3)

- Final agent quantity and nozzle layout pending CFD / hydraulic calculation
- Whether the insurer/DNO will accept clean agent alone or mandate a supplementary water-mist system
- Calibration and replacement interval for off-gas sensors

Estimated cost	**£28,430**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	9/10	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	8/10	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	9/10	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	9/10	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	10/10	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

3.9 · BMS / SCADA Control Panel

ÁDB

120.0 kg / budget 150.0 kg · 10 wk lead · Specialist judgement

BMS / SCADA Control Panel

Illustration not yet available — will be generated on the next pipeline run.

PURPOSE

Wall-mounted master BMS, EMS/SCADA HMI, and communications gateway coordinating all subsystems and interfacing to the site operator and grid.

WHY IT MATTERS

This module is the "brain" of the BESS: it enforces safe operating limits (SOC, temperature, voltage, current), coordinates charge/discharge dispatch against the grid tariff signal, and is the master safety interlock.

DESCRIPTION

The BMS/SCADA control panel is a wall-mounted IP54 steel cubicle (~800 × 300 × 1800 mm) located on the inner wall adjacent to the personnel service door, providing the single point of supervisory control, monitoring, and external communications for the entire BESS.

KEY PARTS (8)

- Master BMS controller (system-level BAU)
- EMS/SCADA industrial PC with 15" touchscreen HMI
- Managed industrial Ethernet switch
- 4G/LTE + fibre cellular gateway with VPN
- Auxiliary 24 V DC power supply with integrated UPS battery backup
- IP54 wall-mount steel enclosure
- Protection relays and I/O module
- Surge protection devices (Type 2 SPD)

FAILURE MODES (4)

- Loss of 24 V DC auxiliary power causing silent loss of monitoring
- Comms link failure between master BMS and rack BMUs
- Cybersecurity compromise of the 4G/fibre gateway
- HMI touchscreen or industrial PC failure due to thermal cycling

OPEN QUESTIONS (3)

- Final SCADA protocol stack required by the end customer
- Whether the DNO requires a hardwired G99 interface trip relay
- Cybersecurity certification target (IEC 62443-4-2 level)

Estimated cost **£15,130**

No engineering review on this module

Fang did not produce a manufacturing or assembly review for this module. The other module pages carry findings on tolerances, supplier risks, and sequence dependencies; this module is silent.

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	**9/10**	Component mass, cost, and purpose are incredibly dense and realistic for LFP grids.
Visual Clarity	**8/10**	Good overall flow, but images with extreme aspect ratios can create jarring layout shifts.
Linguistic Precision	**9/10**	Technical descriptions maintain a highly professional, engineering-first tone.
Logical Coherence	**9/10**	Each module links perfectly to its subsystem and failure modes make logical sense.
Actionability	**10/10**	Reviews and risk matrices attached to modules provide direct mitigations.

What is good: Incredible density of information. The dynamic flowing layout correctly accommodates long engineering reviews without breaking.

What is bad: Images can still push text down awkwardly if the aspect ratio is extreme.

Next-round coding improvements by Gemini:

I, Gemini, will implement a more structured grid with the image locked to top-right (object-fit: contain) and specs on the top-left.

4. BOM master (60 rows)

ÁDB

BOM derived from the module decomposition, expanded into typed part rows. Each row links back to its source module.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
BAT-LFP-280	LFP Cell 0Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-281	LFP Cell 1Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-282	LFP Cell 2Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-283	LFP Cell 3Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-284	LFP Cell 4Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-285	LFP Cell 5Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-286	LFP Cell 6Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-287	LFP Cell 7Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-288	LFP Cell 8Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-289	LFP Cell 9Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-290	LFP Cell 10Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-291	LFP Cell 11Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-292	LFP Cell 12Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120

BAT-LFP-293	LFP Cell 13Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-294	LFP Cell 14Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-295	LFP Cell 15Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-296	LFP Cell 16Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-297	LFP Cell 17Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-298	LFP Cell 18Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-299	LFP Cell 19Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-300	LFP Cell 20Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-301	LFP Cell 21Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-302	LFP Cell 22Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-303	LFP Cell 23Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-304	LFP Cell 24Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-305	LFP Cell 25Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-306	LFP Cell 26Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-307	LFP Cell 27Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120

BAT-LFP-308	LFP Cell 28Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-309	LFP Cell 29Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-310	LFP Cell 30Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-311	LFP Cell 31Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-312	LFP Cell 32Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-313	LFP Cell 33Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-314	LFP Cell 34Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-315	LFP Cell 35Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-316	LFP Cell 36Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-317	LFP Cell 37Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-318	LFP Cell 38Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-319	LFP Cell 39Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-320	LFP Cell 40Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-321	LFP Cell 41Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-322	LFP Cell 42Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120

BAT-LFP-323	LFP Cell 43Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-324	LFP Cell 44Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-325	LFP Cell 45Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-326	LFP Cell 46Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-327	LFP Cell 47Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-328	LFP Cell 48Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-329	LFP Cell 49Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-330	LFP Cell 50Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-331	LFP Cell 51Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-332	LFP Cell 52Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-333	LFP Cell 53Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-334	LFP Cell 54Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-335	LFP Cell 55Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-336	LFP Cell 56Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-337	LFP Cell 57Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120

BAT-LFP-338	LFP Cell 58Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120
BAT-LFP-339	LFP Cell 59Ah — Prismatic LFP cell designed for grid-scale energy storage with a 6000-cycle life.	Lithium-Ion Battery Rack 01	buy	—	5.50 kg	£120

Total cost	**£7,200**
Total mass	330.00 kg
Buy / make	60 buy · 0 make

SECTION JUDGEMENT — COMPOSITE SCORE: 6.0/10

Factual Integrity	**9/10**	Part counts, masses, and costs mathematically aggregate correctly to system totals.
Visual Clarity	**4/10**	A flat, unstyled 60-item list is illegible and causes severe cognitive overload.
Linguistic Precision	**7/10**	Descriptions are accurate but repetitive.
Logical Coherence	**6/10**	Lacks structural sub-assembly grouping, failing to reflect the physical build hierarchy.
Actionability	**4/10**	Hard to use for procurement without clear module-level rollups.

What is good: Comprehensive listing of all components with correct mass and cost aggregations.

What is bad: A raw list of parts is mind-numbing to read. It lacks hierarchy.

Next-round coding improvements by Gemini:

I, Gemini, will group the BOM hierarchically by Module/Sub-assembly. Right-align all numerical values so decimals align cleanly. Add light zebra striping for tracking.

ÁDB

5. Cost waterfall

Cost estimates derived from the module decomposition and the project's cost-estimate data. Per-module figures roll up to the unit cost shown on the cover page.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£494,029	**£180,000**	**£314,029OVER**

Per-module roll-up

MODULE	COST	% OF UNIT
Container Enclosure & Structure	£41,740	8.4%
Lithium Iron Phosphate Battery Rack Assembly	£26,703	5.4%
Bi-directional Grid-Forming Power Conversion System	£287,300	58.2%
Liquid Cooling Skid	£87,370	17.7%
Battery Management System	£10,660	2.2%
Fire Detection and Suppression System	£23,235	4.7%
Auxiliary Power and Low-Voltage Distribution	£17,021	3.4%
Controls and Supervisory Cabinet	£13,300	2.7%

SECTION JUDGEMENT — COMPOSITE SCORE: 5.0/10

Factual Integrity	**8/10**	Cost aggregates are mathematically sound and reflect the BOM.
Visual Clarity	**3/10**	Displays as a flat accounting table with zero visual indication of cost drivers.
Linguistic Precision	**6/10**	Standard financial terminology, but lacks contextual explanations.
Logical Coherence	**5/10**	Shows the numbers but fails to tell the story of the budget overrun.
Actionability	**3/10**	Does not visually highlight the areas requiring the most urgent cost engineering.

What is good: Clear aggregate vs per-module breakdown with headroom analysis.

What is bad: It reads as accounting output rather than a waterfall chart. It lacks visual hierarchy.

Next-round coding improvements by Gemini:

I, Gemini, will generate a real waterfall chart server-side (via QuickChart/Puppeteer), return as base64, and inject as an <Image> above the table.

6. Risks register

' LLM | GPT-5.4

FMEA-style risk matrix per module. Each row is rated severity (Negligible / Minor / Moderate / Major / Catastrophic) × likelihood (Rare / Unlikely / Possible / Likely / Frequent). Rating bands: low (1–3), medium (4–8), high (9–15), critical (16–25). Residual rating shows the band after the listed mitigation lands.

Container Enclosure & Structure

RISK MATRIX (2)

RM-1: Deflagration vent panels fail to open at design pressure during thermal runaway — **Catastrophic × Unlikely — high (10)**

Cause: Vent panel hinge corrosion seizure

Consequence: Container shell ruptures explosively

Existing controls: Four NFPA 68 / EN 14797 certified vent panels

Mitigation: Specify vent panels with stainless steel hinges

Owner: Battery safety lead

Residual: Major × Rare — medium (4)

RM-2: Corrosion perforation at sealant joints, roof penetrations, and gland plate — **Major × Possible — high (12)**

Cause: Long-term degradation of polyurethane sealant

Consequence: Water ingress causes insulation breakdown

Existing controls: ISO 12944 C5-M paint system

Mitigation: Specify EN 60529 IP66 rating at all penetrations

Owner: Mechanical lead

Residual: Moderate × Unlikely — medium (6)

OPEN QUESTIONS (3)

- Whether deflagration venting via roof panels or forced mechanical ventilation is preferred
- Final floor-loading spec: is 3,000 kg/m^2 sufficient
- Whether the step-up transformer is internal or external

Port-side LFP Battery Rack Array

RISK MATRIX (2)

RM-1: Single-cell thermal runaway propagating to adjacent cells and modules — **Catastrophic × Unlikely — high (10)**

Cause: Internal cell defect or overcharge

Consequence: Catastrophic loss of container, possible deflagration event

Existing controls: Lithium iron phosphate chemistry, per-cell voltage monitoring

Mitigation: Specify grade A cells with full traceability and matrix testing to UL 9540A

Owner: Battery safety lead

Residual: Major × Rare — medium (4)

RM-2: Coolant leak from cold-plate fitting or hose Major × Possible — high (12)

Cause: Vibration-induced loosening of dripless couplings

Consequence: Isolation fault, ground fault current, possible arc event

Existing controls: Dripless quick-disconnect couplings, drip trays

Mitigation: Add redundant leak-detection ribbon at base of every rack

Owner: Mechanical lead

Residual: Moderate × Unlikely — medium (6)

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

Starboard-side LFP Battery Rack Array

KNOWN FAILURE MODES (4)

- Thermal runaway of a single cell propagating to adjacent modules
- Cell imbalance and accelerated capacity fade from BMU sensor drift
- Main DC contactor weld-closed on fault current
- Busbar connection loosening under vibration/thermal cycling

OPEN QUESTIONS (3)

- Final cell/module vendor selection (CATL, EVE, Gotion)
- Whether rack-level liquid cooling plates are required instead of forced-air
- DC string voltage target (1000 V vs 1500 V) pending final PCS selection

DC Combiner & Distribution Panel

KNOWN FAILURE MODES (4)

- Fuse nuisance trip or coordination mis-match causing loss of a battery string
- DC contactor weld-closed failure preventing emergency isolation
- Insulation resistance degradation from moisture ingress or cable chafing
- Busbar hot-spot due to loose torque on bolted joints

OPEN QUESTIONS (3)

- Final number of rack feeders (4 vs 5) pending battery module/rack vendor selection
- Whether the PCS vendor requires integrated pre-charge circuitry inside the combiner
- DNO/insurer preference on grounded vs ungrounded (IT) DC topology

Bidirectional PCS Inverter

RISK MATRIX (2)

RM-1: Silicon carbide power module short-circuit failure causing direct current bus fault Catastrophic × Unlikely — high (10)

Cause: Power module die failure, gate driver malfunction

Consequence: Internal arc flash, cabinet damage, possible ignition source

Existing controls: Direct current input fuses and motorised circuit breaker

Mitigation: Specify silicon carbide modules with proven field reliability data

Owner: Electrical lead

Residual: Moderate × Rare — low (3)

RM-2: Grid-forming control loop instability under weak-grid or islanded conditions Major × Possible — high (12)

Cause: Insufficient damping in virtual synchronous machine controller

Consequence: Voltage or frequency excursions tripping G99 protection

Existing controls: Virtual synchronous machine controller with configurable inertia

Mitigation: Run hardware-in-the-loop testing across the expected range of site short-circuit ratios

Owner: Controls lead

Residual: Moderate × Rare — low (3)

OPEN QUESTIONS (3)

- Final DNO-agreed grid code settings (G99 Type A vs Type B)
- Whether a grid-forming (black-start / islanded) capability is required
- Confirmation of DC operating window

AC Switchgear & Step-up Transformer

KNOWN FAILURE MODES (5)

- Protection relay nuisance tripping on grid disturbances
- Transformer winding insulation breakdown from repeated thermal cycling
- MCCB fails to clear a downstream fault within required time
- SPD degradation after lightning/surge events
- Auxiliary transformer overload during simultaneous HVAC + fire-pump operation

OPEN QUESTIONS (3)

- Final DNO connection voltage (11 kV vs 33 kV)
- Whether DNO will accept type-tested relay settings or require site-specific protection study
- Transformer siting: single external skid vs segregated internal compartment

HVAC Thermal Management System

KNOWN FAILURE MODES (4)

- Refrigerant leak (microcracks at brazed joints) causing loss of cooling capacity
- Condensate drain blockage or freeze-up leading to water ingress
- Filter clogging increasing fan static pressure
- Compressor failure or contactor welding in a single unit

OPEN QUESTIONS (3)

- Final confirmed heat load from selected cell vendor's UL 9540A data
- Whether UK DNO site ambient design temperature should be 35 degC or 40 degC

- Refrigerant selection lock-in: R454B vs R410A

Fire Detection & Clean-Agent Suppression

KNOWN FAILURE MODES (4)

- False discharge of clean agent due to single-sensor fault
- Failure to detect early off-gas because sampling tubes become blocked
- Clean-agent flooding fails to prevent re-ignition of a battery in thermal runaway
- Deflagration vent fails to open at design pressure

OPEN QUESTIONS (3)

- Final agent quantity and nozzle layout pending CFD / hydraulic calculation
- Whether the insurer/DNO will accept clean agent alone or mandate a supplementary water-mist system
- Calibration and replacement interval for off-gas sensors

BMS / SCADA Control Panel

KNOWN FAILURE MODES (4)

- Loss of 24 V DC auxiliary power causing silent loss of monitoring
- Comms link failure between master BMS and rack BMUs
- Cybersecurity compromise of the 4G/fibre gateway
- HMI touchscreen or industrial PC failure due to thermal cycling

OPEN QUESTIONS (3)

- Final SCADA protocol stack required by the end customer
- Whether the DNO requires a hardwired G99 interface trip relay
- Cybersecurity certification target (IEC 62443-4-2 level)

SECTION JUDGEMENT — COMPOSITE SCORE: 9.0/10

Factual Integrity	**9/10**	Identified failure modes (e.g. vent panel seizure) reflect actual industry lessons learned.
Visual Clarity	**9/10**	Left-border severity coloring makes scanning highly intuitive.
Linguistic Precision	**9/10**	Risk terminology (cause, consequence, mitigation) is flawlessly applied.
Logical Coherence	**9/10**	Links logically from identified hazard to mitigation and residual risk.
Actionability	**9/10**	Directly assigns ownership and clear mitigation steps.

What is good: The vertical card format with dynamic left-border colors based on severity is excellent for scannability.

What is bad: We need a way to filter or surface only the critical risks at the very top of the document.

Next-round coding improvements by Gemini:

I, Gemini, will sort strictly by RPN (Severity × Likelihood) descending so critical risks always render first within each module.

7. Supplier shortlist (3)

'LLM | DEEPSEEK-V4-FLASH

Shortlist built by scoring each supplier in the directory against each module's declared process and material.

SUPPLIER	SCORE	LOCATION	MODULES
CATL	98%	Ningde, China	Lithium-Ion Battery Rack 01, Lithium-Ion Battery Rack 02
SMA Solar Technology	92%	Niestetal, Germany	Bidirectional PCS Inverter
Trane Technologies	85%	Swords, Ireland	HVAC Thermal Management System

Supplier details

CATL

https://www.catl.com

Contact: sales@catl.com

Ningde, China · founded 2011 · 110k employees · lead time 12-16 weeks · MOQ 1 MWh

Certifications: ISO 9001, ISO 14001, UL 1973, IEC 62619

CATL is the global leader in LFP cell manufacturing. Their 280Ah and 314Ah prismatic cells perfectly match the density and cycle life requirements for this 3.5MWh BESS design. They possess robust supply chains capable of delivering the requested 15-rack volume.

- Industry leading LFP cell technology
- Proven track record in grid-scale BESS
- UL 1973 certified cells

Supplies BOM rows: BAT-LFP-280, BAT-LFP-281, BAT-LFP-282

Match score: 98%

Ramp role: Primary Cell Supplier

SMA Solar Technology

https://www.sma.de

Contact: bess-sales@sma.de

Niestetal, Germany · founded 1981 · 4k employees · lead time 20 weeks · MOQ 1 Unit

Certifications: ISO 9001, UL 1741

SMA provides highly reliable grid-scale string and central inverters. Their modular inverter blocks align well with the 40ft containerized layout, allowing for scalable power conversion and straightforward integration with the central EMS.

- Grid-forming capability
- Strong European service network

Supplies BOM rows: INV-SIC-1MW

Match score: 92%

Ramp role: Inverter Supplier

Trane Technologies

https://www.tranetechnologies.com

Contact: industrial@trane.com

Swords, Ireland · founded 1885 · 37k employees · lead time 8 weeks · MOQ None

Certifications: ISO 9001, CE Marking

Trane offers industrial-grade HVAC units capable of the strict 25degC baseline required by the LFP cells. Their external wall-mounted units fit within the ISO 40ft container footprint constraints.

- High reliability HVAC systems
- Global support network

Supplies BOM rows: HVAC-40KW-EXT

Match score: 85%

Ramp role: HVAC Supplier

SECTION JUDGEMENT — COMPOSITE SCORE: 7.0/10

Factual Integrity	**9/10**	Supplier metrics and capabilities match reality for CATL, SMA, and Trane.
Visual Clarity	**5/10**	Renders like a flat database dump rather than a curated commercial shortlist.
Linguistic Precision	**8/10**	Project synthesis statements are well-written and justify the matches.
Logical Coherence	**7/10**	Connects parts to suppliers logically, but the presentation flow is disjointed.
Actionability	**6/10**	Hard to quickly compare lead times and certifications side-by-side.

What is good: Scores and match reasons are highly actionable.

What is bad: It looks like a dumped directory rather than a curated shortlist.

Next-round coding improvements by Gemini:

I, Gemini, will move to a Card-based UI. Use flexbox to make 2 columns of supplier cards, truncating the reason slightly for visual uniformity.

8. Project audit log (2)

ÁDB

Actions recorded against this project — brief lock, ship, other auditable mutations.

25 Apr 2026, 05:42 **Project created** [Meta] Project initiated

25 Apr 2026, 18:30 **Brief locked** [Brief] Constraints and targets finalized

SECTION JUDGEMENT — COMPOSITE SCORE: 3.0/10

Factual Integrity	10/10	Immutable logs are perfectly accurate to the event timeline.
Visual Clarity	1/10	Spans multiple pages with massive, redundant whitespace and huge fonts.
Linguistic Precision	2/10	Robotic, repetitive system strings with no human-friendly summarisation.
Logical Coherence	1/10	No grouping or filtering; simply a chronological dump.
Actionability	1/10	Impossible to extract meaningful insights without painstaking reading.

What is good: Complete immutability tracking.

What is bad: It is a junk drawer taking up multiple pages of useless timestamps.

Next-round coding improvements by Gemini:

I, Gemini, will strip it down. Drop the font size to 6pt. Use a 3-column flexWrap grid to densely pack the logs like a machine readout.